

UNIVERZITA KOMENSKÉHO V BRATISLAVE
FAKULTA MATEMATIKY, FYZIKY A INFORMATIKY

AUTOMATIC CITATION INTEGRITY
VERIFICATION FOR LLM-ASSISTED ACADEMIC
WRITING
DIPLOMA THESIS



2027

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Študijný program: Aplikovaná Informatika
Študijný odbor: Informatika
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THESIS ASSIGNMENT

Name and Surname:	Bc. Vladimír Tischler
Study programme:	Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study:	Computer Science
Type of Thesis:	Diploma Thesis
Language of Thesis:	English
Secondary language:	Slovak
Title:	Automatic Citation Integrity Verification for LLM-Assisted Academic Writing
Annotation:	Large language models can produce plausible-looking bibliographies and in-text citations that may be partially or fully fabricated, contain metadata errors, or fail to support the surrounding claim. This thesis aims to design and evaluate a pipeline that extracts and normalizes reference strings and in-text citation anchors, resolves them against scholarly metadata sources to detect non-existent or inconsistent works, and assesses whether the cited work plausibly supports the local citation context (at sentence/paragraph level). The intended outcome is a practical verification component that flags high-risk citations with interpretable error categories (e.g., non-resolving reference, mismatched venue/year/authors, weak topical alignment, unsupported claim-to-citation link).
Aim:	The goals of the thesis include (but are not limited to): Curating an evaluation set of academic passages with citations, including LLM-generated references, with labels for existence, metadata correctness, and context support. Benchmarking reference parsing and normalization approaches for noisy, style-diverse bibliographic strings, emphasizing robustness to incomplete metadata. Developing a citation-resolution module (identifier recovery + fuzzy matching) that aims to separate “non-existent” from “real-but-misformatted/mismatched” references. Prototyping a lightweight “citation support” check that aims to score whether the cited work aligns with the claim made near the citation, and analyzing failure modes. Evaluating operational trade-offs (precision at high recall, latency/cost, domain shift) and proposing reporting thresholds suitable for editorial/educational use.
Literature:	Walters, W. H. “Fabrication and errors in the bibliographic citations generated by ChatGPT.” <i>Scientific Reports</i> (2023). https://www.nature.com/articles/s41598-023-41032-5 Guenci, M., Heibi, I., Parravicini, C., Peroni, S., Soricetti, M. “A pipeline for matching bibliographic references with incomplete metadata: experiments with Crossref and OpenCitations.” <i>arXiv:2511.18408</i> (2025). https://arxiv.org/abs/2511.18408 Alvarez, C., et al. “Zero-shot Scientific Claim Verification Using LLMs and Citation Text” <i>ACL Workshop</i> (2024). https://aclanthology.org/2024.sdp-1.25/ Press, Ori, Andreas Hochlehnert, Ameya Prabhu, Vishaal Udandarao, Ofir Press, and Matthias Bethge. “CiteME: Can Language Models Accurately Cite



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Scientific Claims?" arXiv preprint arXiv:2407.12861 (2024). <https://arxiv.org/abs/2407.12861>

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Statutory Declaration

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Abstract

Keywords:

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Bibliography

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